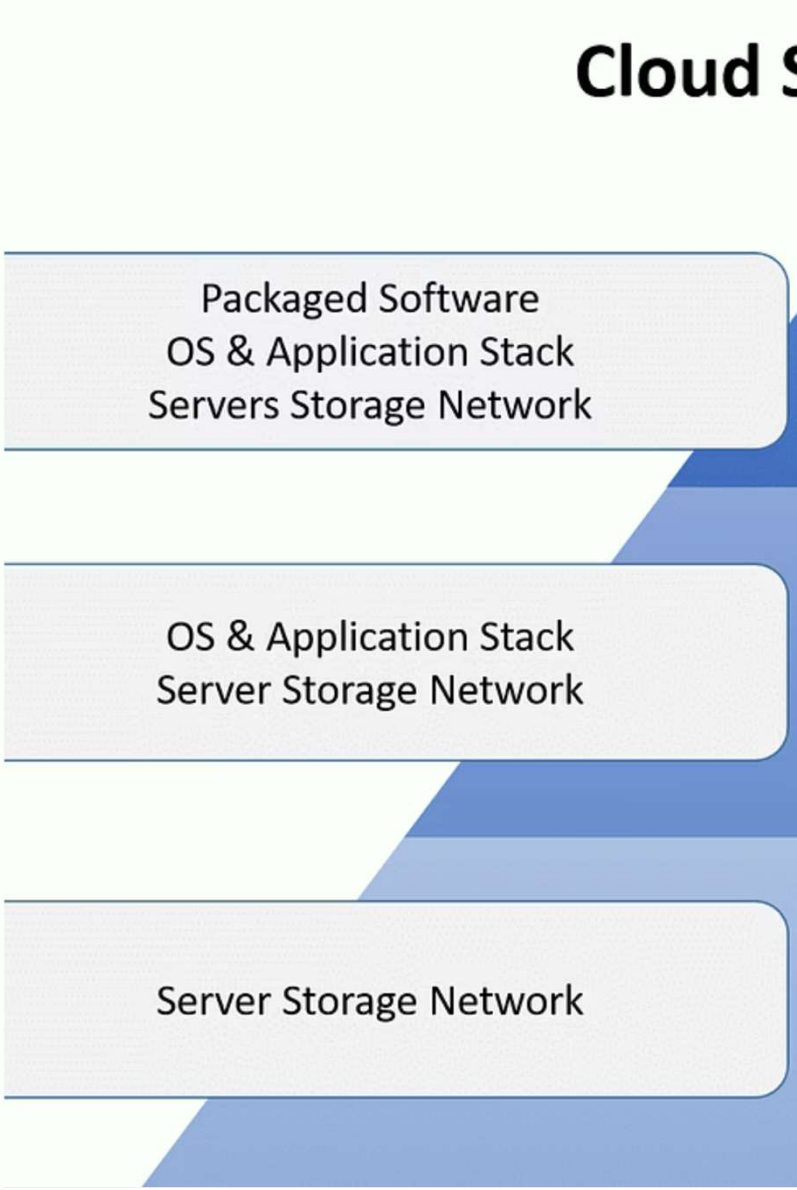
AWS Application Integration Services: SQS, SNS, and EventBridge

Discover how Amazon's powerful messaging and event services can help you build resilient, decoupled applications in the cloud.



Introduction

Application Integration

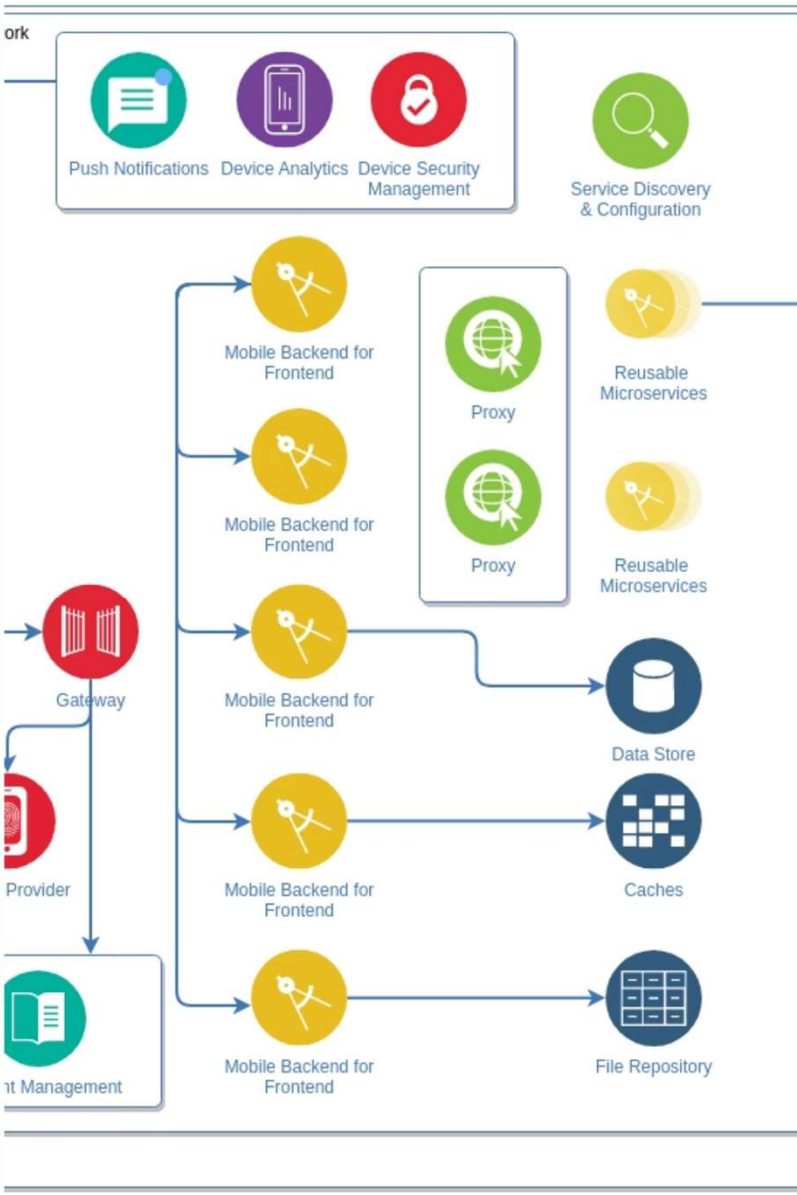
Application integration connects distributed systems, enabling them to communicate effectively while remaining independent.

Decoupling Benefits

Decoupling services improves resilience, scalability, and maintainability of cloud applications.

Today's Focus

- Amazon SQS - Reliable messaging queues
- Amazon SNS - Publish/subscribe messaging
- Amazon EventBridge - Event-driven architecture



Amazon SQS Overview

Amazon Simple Queue Service (SQS) is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications.

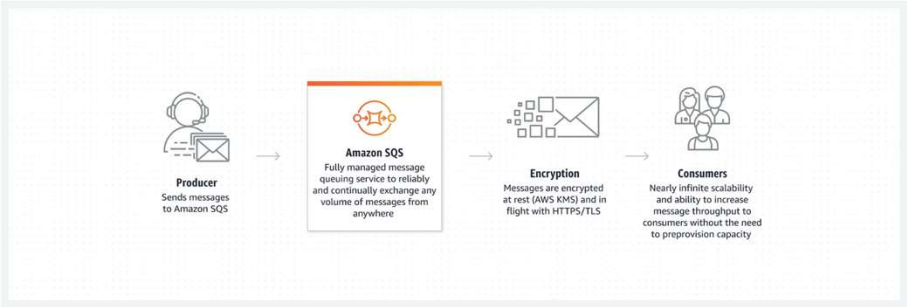
Queue Types:

Standard Queue

High throughput, at-least-once delivery, best-effort ordering

FIFO Queue

First-In-First-Out delivery, exactly-once processing



Common Use Cases:

- Order processing systems
- Background job management
- Work distribution across multiple servers

SQS Deep Dive

Send

Producer sends message to queue

Store

Message stored redundantly across servers

Receive

Consumer polls and processes message

Delete

Message deleted after successful processing

Key Features:

- **Visibility Timeout:** Prevents double processing
- **Dead-Letter Queue (DLQ):** Captures failed messages
- **Security:** IAM policies, SSE encryption
- **Long Polling:** Reduces empty responses

```
aws sqs create-queue --queue-name MyQueueaws sqs send-message --queue-url --message-body "Hello World"
```

Amazon SNS Overview

Amazon Simple Notification Service (SNS) is a fully managed **publish/subscribe** messaging service for both application-to-application (A2A) and application-to-person (A2P) communication.

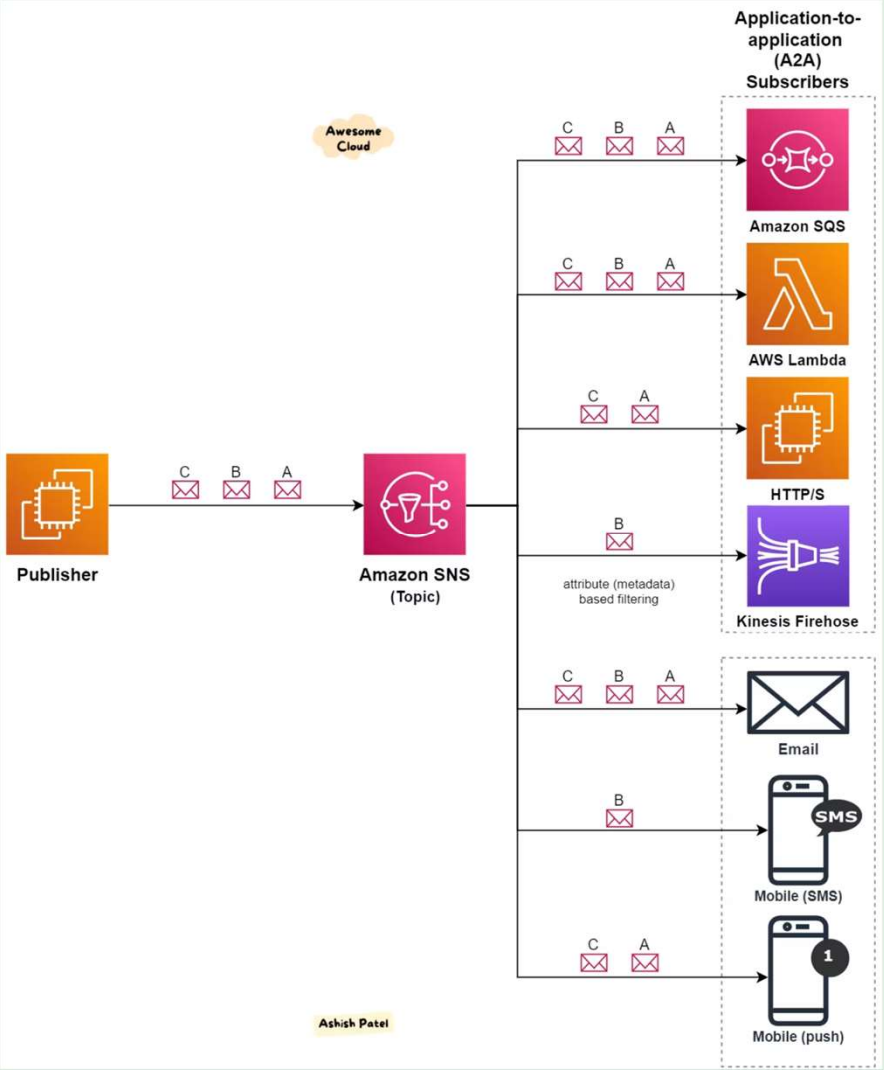
Key Capabilities:

- Publishers send messages to a topic
- Multiple subscribers receive messages
- Supports various subscription protocols

Subscription Types:

- Email and SMS
- HTTP(S) endpoints
- AWS Lambda functions
- SQS queues
- Mobile push notifications

Primary Use Cases: Alerts, notifications, fan-out messaging patterns



SNS Deep Dive

1

Topic Types

- **Standard:** Best-effort ordering, at-least-once delivery
- **FIFO:** Strict ordering, exactly-once delivery (SQS only)

2

Message Filtering

Filter messages using attributes to deliver only relevant messages to specific subscribers

3

Delivery Retries & DLQs

Configurable retry policy with Dead-Letter Queues for SQS subscribers

4

Security

Topic policies, encryption in transit and at rest, access control

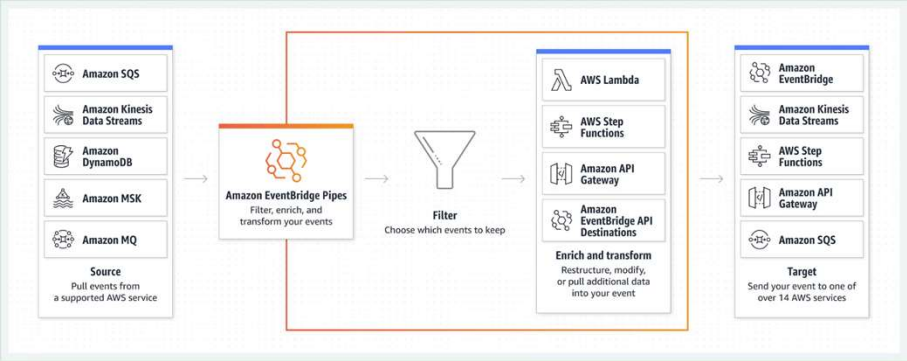
```
aws sns create-topic --name MyTopicaws sns subscribe --topic-arn --protocol email --notification-endpoint
example@example.com
```

Amazon EventBridge Overview

Amazon EventBridge is a serverless event bus service that makes it easy to connect applications together using data from your own applications, integrated SaaS applications, and AWS services.

Event Sources:

- AWS services (EC2, S3, etc.)
- SaaS applications (Zendesk, DataDog, etc.)
- Custom applications



Event Targets:

- AWS Lambda functions
- SQS queues and SNS topics
- Step Functions state machines
- API Gateway endpoints
- Many other AWS services

Primary Use Cases: Event-driven microservices, workflow automation, application integration

EventBridge Deep Dive

Event Rules

Rules match events using patterns and route them to targets. You can create rules that react to events from AWS services or your own custom events.

```
{"source": ["aws.ec2"], "detail-type": ["EC2 Instance State-change Notification"]}
```

Schema Registry

Discover, create, and manage schemas for events. Generate code bindings for popular programming languages.

Archive & Replay

Store events for a specified period and replay them to new or updated targets for testing or recovery.

Security

IAM permissions control who can publish events and create rules. Resource-based policies for cross-account access.

```
aws events put-rule --name MyRule --event-pattern '{"source": ["aws.ec2"]}'
```

Service Comparison

Feature	SQS	SNS	EventBridge
Pattern	Queue	Pub/Sub	Event Bus
Delivery	Pull	Push	Push
Order	FIFO optional	FIFO optional	N/A
Filtering	No	Yes	Yes
Fan-out	No	Yes	Yes

Each service has distinct capabilities that make it suitable for different integration patterns and use cases.

Integration Example

EC2 Auto-Scaling Notification System

Step 1: Event Trigger

EC2 instance launches and sends state-change event to EventBridge

Step 2: Event Routing

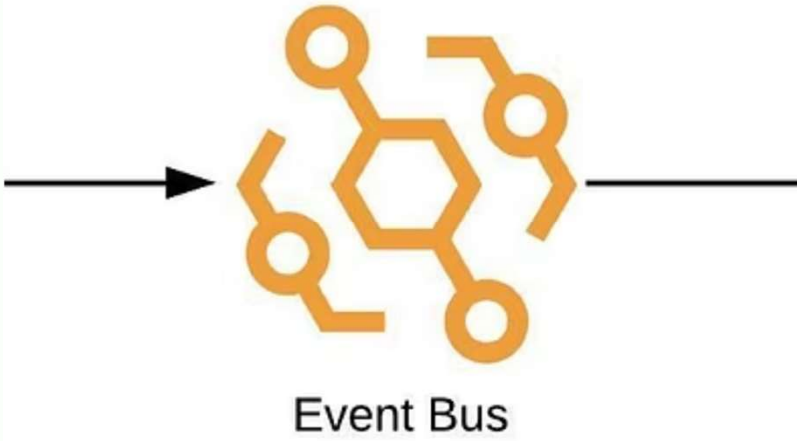
EventBridge rule matches the event and routes it to an SNS topic

Step 3: Notification Distribution

SNS topic fans out to multiple subscribers: email alerts and SQS queues

Step 4: Asynchronous Processing

Different services process messages from SQS queues independently



Summary

SQS

Reliable, scalable queuing for message buffering and decoupling components

SNS

Flexible pub/sub messaging for broadcasting to multiple endpoints

EventBridge

Central event routing for building event-driven architectures

Best Practices:

- Combine services for robust integration patterns
- Use SQS for buffering and retries
- Use SNS for fan-out messaging
- Use EventBridge for complex event routing

These services form the backbone of resilient, scalable, and loosely coupled cloud architectures.